

Phys 115A HW 5

Zih-Yu Hsieh

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Question 1. Odd Wells (Griffiths 2.29)

Let's go back to the odd case of the finite square well:

- 1. Write down the general form of the odd bound-state wavefunctions in the three regions $x < -a$, $-a \leq x \leq a$, and $x > a$.*
- 2. Find the transcendental equation for the allowed energies. Solve it graphically.*
- 3. What is the energy spectrum of bound states in the limiting cases of a wide, deep well and a shallow, narrow well (in the sense that Griffiths describes in Section 2.6 around Eq. 2.157)? Do we always find an odd bound state?*

Question 2. *Delta limit (Griffiths 2.31)*

As we discussed in class, a Dirac delta function can be thought of as a limiting case of a finite square well, with the added condition that the area is fixed to be 1 as the height goes to infinity and the width goes to zero. Show that the delta function well is a “weak” potential in the sense that $z_0 \rightarrow 0$. Determine the bound state energy for the delta-function potential by treating it as the limit of the finite square well. Check that your answer is consistent with Eq. 2.132. Also show that Eq. 2.172 reduces to Eq. 2.144 in the appropriate limit.

Question 3. *The Square Barrier (Griffiths 2.33)*

Let's work out the transmission coefficient for a rectangular barrier defined by

$$V(x) = \begin{cases} V_0 & \text{for } -a \leq x \leq a, \\ 0 & \text{for } |x| > a \end{cases}$$

for $V_0 > 0$. There are three possible cases of interest: $E < V_0$; $E = V_0$; and $E > V_0$, and the wavefunction inside the barrier is different in each case.

1. Find T^{-1} for $E < V_0$
2. Find T^{-1} for $E = V_0$
3. Find T^{-1} for $E > V_0$ (Hint: This one is easily related to the finite square well.)

Partial answer: for $E < V_0$, $T^{-1} = 1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2 \left(\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} \right)$

Question 4. *Mass on (circular) string (Griffiths 2.46)*

Imagine a bead of mass m that slides frictionlessly around a circular wire ring of circumference L . (This is like a free particle except that $\psi(x + L) = \psi(x)$.) Find the stationary states (with appropriate normalization) and the corresponding allowed energies. Note that there are (with one exception) two independent solutions for each energy E_n —corresponding to clockwise and counter-clockwise circulation; call them $\psi_n^+(x)$ and $\psi_n^-(x)$. How do you account for this degeneracy, in view of the theorem of problem 2.44 (why does the theorem fail in this case)? Hint: you worked through problem 2.44 in discussion section.

Question 5. *The Transfer Matrix (Griffiths 2.54)*

The transfer matrix. The S-matrix (Problem 2.53) tells you the outgoing amplitudes (B and F) in terms of the incoming amplitudes (A and G) – Equation 2.180. For some purposes it is more convenient to work with the transfer matrix, M , which gives you the amplitudes to the right of the potential (F and G) in terms of those to the left (A and B):

$$(F, G)^T = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} (A, B)^T$$

- Find the four elements of the M -matrix, in terms of the elements of the S -matrix, and vice versa. Express R_l , T_l , R_r , and T_r (Equations 2.176 and 2.177) in terms of elements of the M -matrix.*
- Suppose you have a potential consisting of two isolated pieces (Figure 2.23). Show that the M -matrix for the combination is the product $M = M_2 M_1$.*
- Construct the M -matrix for scattering from a single delta-function potential at point a : $V(x) = -\alpha\delta(x - a)$*
- By the method of part (b), find the M -matrix for scattering from the double delta function $V(x) = -\alpha[\delta(x + a) + \delta(x - a)]$ What is the transmission coefficient for this potential?*

Question 6. *Linear Algebra Bootcamp I: Vector spaces (modified Griffiths A.3)*

1. Prove that the components of a vector in a given basis are unique.
2. Consider the set of all polynomials in x of degree $\leq N$ with complex coefficients. Does this set comprise a vector space, where the polynomials in x are the “vectors”? If so, suggest a convenient basis. What is the dimension of this basis? If not, where does it fall short?
3. Does the set of odd polynomials in x constitute a vector space? Why or why not?

Proof.

1. Let V be a $\mathbb{K}$ -vector space (where $\mathbb{K}$ denotes the field). Let I denote an index set, where $\{v_i\}_{i \in I} \subset V$ is a basis of V . Which, the list is linearly independent, meaning that if $\sum_{i \in I} c_i v_i = 0$ (where each $c_i \in \mathbb{K}$), then it implies $c_i = 0$ for all $i \in I$.

As a result, given any $v \in V$, since $\{v_i\}_{i \in I}$ spans V , $v = \sum_{i \in I} c_i v_i$ for some list of coefficients $\{c_i\}_{i \in I} \subset \mathbb{K}$. Now, for uniqueness suppose the two lists of coefficients $\{c_i\}_{i \in I}, \{d_i\}_{i \in I} \subset \mathbb{K}$ satisfies $v = \sum_{i \in I} c_i v_i = \sum_{i \in I} d_i v_i$, then $0 = v - v = \sum_{i \in I} (c_i - d_i) v_i$, with the claim of linear independence one must have $c_i - d_i = 0$, or $c_i = d_i$ for all $i \in I$.

Hence, this shows that the component (i.e. the coefficient of each v_i for v) must be unique.

2. Let $\mathcal{P}_N(\mathbb{C})$ denotes all degree $\leq N$ polynomials with complex coefficients, to show it's a vector space it suffices to show given any $f, g \in \mathcal{P}_N(\mathbb{C})$, arbitrary linear combination $af + bg \in \mathcal{P}_N(\mathbb{C})$ (since choose $a, b = 0 \in \mathbb{C}$, this shows $0 \in \mathcal{P}_N(\mathbb{C})$, the set has a zero vector; choose $a, b = 1 \in \mathbb{C}$, it shows $f + g \in \mathcal{P}_N(\mathbb{C})$, which is closed under addition; also, choose $b = 0$ and arbitrary $a \in \mathbb{C}$, this shows $af \in \mathcal{P}_N(\mathbb{C})$, so it's closed under scalar multiplication).

For any $f, g \in \mathcal{P}_N(\mathbb{C})$, there exists $\{a_j\}_{j=0}^N, \{b_j\}_{j=0}^N \subset \mathbb{C}$, such that $f(z) = \sum_{j=0}^N a_j z^j$, and $g(z) = \sum_{j=0}^N b_j z^j$. Which, given arbitrary $k, l \in \mathbb{C}$, we have the following:

$$k \cdot f(z) + l \cdot g(z) = k \cdot \sum_{i=0}^N a_i z^i + l \cdot \sum_{j=0}^N b_j z^j = \sum_{j=0}^N (k \cdot a_j) z^j + \sum_{j=0}^N (l \cdot b_j) z^j = \sum_{j=0}^N (k \cdot a_j + l \cdot b_j) z^j \quad (1)$$

Since each $j \in \{0, 1, \dots, N\}$ satisfies $k \cdot a_j + l \cdot b_j \in \mathbb{C}$ (since $\mathbb{C}$ is a field), then $kf + lg$ is still a polynomial, and it has degree at most N (since it can only have nonzero coefficients for degree $\leq N$). Hence, $kf + lg \in \mathcal{P}_N(\mathbb{C})$, this shows that $\mathcal{P}_N(\mathbb{C})$ is a vector space.

A convenient basis is $\{z^j\}_{j=0}^N$ (since a polynomial is 0 iff all the coefficients of each z^j is 0, so $\{z^j\}$ if contained in the space, is linearly-independent; and, since $\{z^j\}_{j=0}^N \subset \mathcal{P}_N(\mathbb{C})$, together with the fact that all polynomials is a linear combination of these, the given list does form a basis).

3. Let L denotes all odd polynomials in x (say over $\mathbb{K}$ a field), then any $f \in L$ satisfies $f(-x) = -f(x)$.

Again, it suffices to show arbitrary finite linear combination still belongs to the space: Given any $f, g \in L$, and any scalar $a, b \in \mathbb{K}$, we have the following:

$$(af + bg)(-x) = a \cdot f(-x) + b \cdot g(-x) = a \cdot (-1) \cdot f(x) + b \cdot (-1) \cdot g(x) = -(af + bg)(x) \quad (2)$$

Hence, $af + bg \in L$, showing that L is closed under arbitrary linear combination, hence a vector space.

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Question 7. *Linear Algebra Bootcamp II: Inner products (Griffiths A.5)*

Prove the Schwartz inequality $|\langle \alpha | \beta \rangle|^2 \leq \langle \alpha | \alpha \rangle \langle \beta | \beta \rangle$.

Hint: Let $|\delta\rangle = |\beta\rangle - \frac{\langle \alpha | \beta \rangle}{\langle \alpha | \alpha \rangle} |\alpha\rangle$ and use $\langle \delta | \delta \rangle \geq 0$.

Proof. Given $|\delta\rangle = |\beta\rangle - \frac{\langle \alpha | \beta \rangle}{\langle \alpha | \alpha \rangle} |\alpha\rangle$, we have the following inequality:

$$0 \leq \langle \delta | \delta \rangle = \langle \beta | \beta \rangle + \left| \frac{\langle \alpha | \beta \rangle}{\langle \alpha | \alpha \rangle} \right|^2 \langle \alpha | \alpha \rangle - \frac{\langle \alpha | \beta \rangle}{\langle \alpha | \alpha \rangle} \langle \beta | \alpha \rangle - \frac{\langle \alpha | \beta \rangle^*}{\langle \alpha | \alpha \rangle} \langle \alpha | \beta \rangle \quad (3)$$

$$\Rightarrow 0 \leq \langle \beta | \beta \rangle + \frac{|\langle \alpha | \beta \rangle|^2}{\langle \alpha | \alpha \rangle} - 2 \frac{|\langle \alpha | \beta \rangle|^2}{\langle \alpha | \alpha \rangle} \quad (4)$$

$$\Rightarrow \frac{|\langle \alpha | \beta \rangle|^2}{\langle \alpha | \alpha \rangle} \leq \langle \beta | \beta \rangle \quad (5)$$

$$\Rightarrow |\langle \alpha | \beta \rangle|^2 \leq \langle \alpha | \alpha \rangle \langle \beta | \beta \rangle \quad (6)$$

□

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Question 8. *Linear Algebra Bootcamp III: Matrices (modified Griffiths A.8)*

Given the matrices $A = \begin{pmatrix} 1 & -1 & i \\ 3 & 0 & 2 \\ -2i & 2i & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 0 & i \\ 0 & 1 & 0 \\ i & 2 & 1 \end{pmatrix}$

compute the following:

1. $A + B$
2. AB
3. $[A, B]$
4. A^T
5. A^*
6. $A^\dagger$
7. $\det(B)$

Proof.

1.

$$A + B = \begin{pmatrix} 4 & -1 & 2i \\ 3 & 1 & 2 \\ -i & 2+2i & 3 \end{pmatrix} \quad (7)$$

2.

$$AB = \begin{pmatrix} 1 & -1 & i \\ 3 & 0 & 2 \\ -2i & 2i & 2 \end{pmatrix} \begin{pmatrix} 3 & 0 & i \\ 0 & 1 & 0 \\ i & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1+2i & 2i \\ 9+2i & 4 & 2+3i \\ -4i & 4+2i & 4 \end{pmatrix} \quad (8)$$

3.

$$[A, B] = AB - BA = \begin{pmatrix} 2 & -1+2i & 2i \\ 9+2i & 4 & 2+3i \\ -4i & 4+2i & 4 \end{pmatrix} - \begin{pmatrix} 3 & 0 & i \\ 0 & 1 & 0 \\ i & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & i \\ 3 & 0 & 2 \\ -2i & 2i & 2 \end{pmatrix} \quad (9)$$

$$= \begin{pmatrix} 2 & -1+2i & 2i \\ 9+2i & 4 & 2+3i \\ -4i & 4+2i & 4 \end{pmatrix} - \begin{pmatrix} 5 & -5 & 5i \\ 3 & 0 & 2 \\ 5-i & i & 5 \end{pmatrix} = \begin{pmatrix} -3 & 4+2i & -3i \\ 6+2i & 4 & 3i \\ -5-3i & 4+i & -1 \end{pmatrix} \quad (10)$$

4.

$$A^T = \begin{pmatrix} 1 & 3 & -2i \\ -1 & 0 & 2i \\ i & 2 & 2 \end{pmatrix} \quad (11)$$

5.

$$A^* = \begin{pmatrix} 1 & -1 & -i \\ 3 & 0 & 2 \\ 2i & -2i & 2 \end{pmatrix} \quad (12)$$

6.

$$A^\dagger = \begin{pmatrix} 1 & 3 & 2i \\ -1 & 0 & -2i \\ -i & 2 & 2 \end{pmatrix} \quad (13)$$

7.

$$\det(B) = -0 \cdot \det \begin{pmatrix} 0 & i \\ 2 & 1 \end{pmatrix} + 1 \cdot \det \begin{pmatrix} 3 & i \\ i & 1 \end{pmatrix} - 0 \cdot \det \begin{pmatrix} 3 & 0 \\ i & 2 \end{pmatrix} = 4 \quad (14)$$

□

Question 9. *Linear Algebra Bootcamp IV: Eigenvectors & Eigenvalues (modified A.19, A.22, A.23)*

1. Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} i & i \\ 0 & i \end{pmatrix}$. Can this matrix be diagonalized?
2. Consider the matrix $B = \begin{pmatrix} i & i \\ i & -1 \end{pmatrix}$. Is it normal? Is it diagonalizable?
3. Show that if two matrices commute in one basis, they commute in any basis.

Proof.

1. Given A an upper-triangular matrix. By linear algebra theory, all its eigenvalues must appear on the diagonal (when it's in upper-triangular form), hence A only has eigenvalue i .

Any of the eigenvector $v \in \mathbb{C}^2$ of A must satisfy $(A - i \cdot I)v = 0$, which $v = (x, y)$ must satisfy the following:

$$0 = (A - i \cdot I)v = \left(\begin{pmatrix} i & i \\ 0 & i \end{pmatrix} - i \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} i \cdot y \\ 0 \end{pmatrix} \quad (15)$$

Hence, $i \cdot y = 0$, implying $y = 0$. So, we must have $v = (x, 0)$, showing the eigenvectors of eigenvalue i are all of the form $v = (x, 0)$ (where $x \in \mathbb{C} \setminus \{0\} = \mathbb{C}^\times$).

However, this matrix is not diagonalizable, since the only eigenspace of A is $E(i) = \text{span}(1, 0) \neq \mathbb{C}^2$, so the eigenvectors don't span the whole $\mathbb{C}^2$, showing A is not diagonalizable.

2. Given $B = \begin{pmatrix} i & i \\ i & -1 \end{pmatrix}$, its conjugate transpose (or adjoint operator / Hermitian conjugate under standard basis) is given by $B^* = \begin{pmatrix} -i & -i \\ -i & -1 \end{pmatrix}$. Which, they satisfy the following:

$$BB^* = \begin{pmatrix} i & i \\ i & -1 \end{pmatrix} \begin{pmatrix} -i & -i \\ -i & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1-i \\ 1+i & 2 \end{pmatrix}, \quad B^*B = \begin{pmatrix} -i & -i \\ -i & -1 \end{pmatrix} \begin{pmatrix} i & i \\ i & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1+i \\ 1-i & 2 \end{pmatrix} \quad (16)$$

Since $BB^* \neq B^*B$, then B is not normal.

However, if consider its characteristic polynomial, it's given as follow:

$$\det(B - \lambda I) = \det \begin{pmatrix} i - \lambda & i \\ i & -1 - \lambda \end{pmatrix} = (i - \lambda)(-1 - \lambda) - i^2 = \lambda^2 + (1 - i)\lambda + (1 - i) \quad (17)$$

Which, if $\det(B - \lambda I) = 0$, since the discriminant of the above quadratic polynomial is $\Delta = (1 - i)^2 - 4(1 - i) = (-3 - i)(1 - i) \neq 0$, this indicates that $\det(B - \lambda I)$ has two distinct roots, hence B has two distinct eigenvalues.

Since distinct eigenvalues must have linearly independent eigenvectors, with two distinct eigenvalues, the two corresponding eigenvectors must be linearly independent. And, with $\mathbb{C}^2$ being 2-dimension (over $\mathbb{C}$), this shows that there exists a basis formed by the eigenvectors of B , hence B is diagonalizable.

3. Given a finite-dimensional $\mathbb{K}$ -vector space V (where $\mathbb{K}$ is a field), and have any linear operator $A, B : V \rightarrow V$. Let α be an ordered basis, and let $\mathcal{M}(A, \alpha), \mathcal{M}(B, \alpha)$ denote the matrix of A, B under ordered basis α . If $\mathcal{M}(A, \alpha)\mathcal{M}(B, \alpha) = \mathcal{M}(B, \alpha)\mathcal{M}(A, \alpha)$ (the matrix representative under basis α commutes), then given any other ordered basis β of V , let T_α^β denote the change of basis matrix from ordered basis β to α (which the inverse $(T_\alpha^\beta)^{-1}$ is the change of basis matrix from α to β), then we have the following:

$$\mathcal{M}(C, \beta) = (T_\alpha^\beta)^{-1} \mathcal{M}(C, \alpha) T_\alpha^\beta \quad (18)$$

Where $C : V \rightarrow V$ is arbitrary linear operator on V , and $\mathcal{M}(C, \beta)$ denotes its matrix representative in basis β .

Hence, we get the following:

$$\mathcal{M}(A, \beta)\mathcal{M}(B, \beta) = ((T_\alpha^\beta)^{-1} \mathcal{M}(A, \alpha) T_\alpha^\beta) ((T_\alpha^\beta)^{-1} \mathcal{M}(B, \alpha) T_\alpha^\beta) \quad (19)$$

$$= (T_\alpha^\beta)^{-1} \mathcal{M}(A, \alpha) \mathcal{M}(B, \alpha) T_\alpha^\beta \quad (20)$$

$$= (T_\alpha^\beta)^{-1} \mathcal{M}(B, \alpha) \mathcal{M}(A, \alpha) T_\alpha^\beta \quad (21)$$

$$= ((T_\alpha^\beta)^{-1} \mathcal{M}(B, \alpha) T_\alpha^\beta) ((T_\alpha^\beta)^{-1} \mathcal{M}(A, \alpha) T_\alpha^\beta) \quad (22)$$

$$= \mathcal{M}(B, \beta) \mathcal{M}(A, \beta) \quad (23)$$

Hence, A, B as linear operators still commute as matrices under basis β . Then, because β as a basis of V is arbitrary, A, B as linear operators have their matrices commute under any basis. (Which comes down to the fact that as linear operators A, B already commutes, so regardless of the matrix representative they should be the same).

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